M1

Research question(s) and target results

Freeze Item	Details
RQ	RQ1. Can the bounded JDK/build-tool success pattern be reproduced?
Source anchor	Section 4.1, Figure 1 for build success/failure across JDK versions; Section 4.4, Table 5/Figure 3 for success by build tool.
Target result	Produce a buildability matrix for 20 projects across JDK 11, 17, 21 and 23, comparing success rates with the original study.
Out of scope	Full 2,500-project study and JDK versions outside **11, 17, 21 and 23**.

Freeze Item	Details
RQ	RQ2. How effective is a simple JDK-switch repair before code changes?
Source anchor	Section 4.2, original RQ2 – projects failing on JDK 23 that successfully build using an earlier JDK.
Target result	Determine how many JDK 23 failures build successfully using JDK 21, 17 or 11 without code changes.
Out of scope	Code changes, refactoring or other repairs beyond switching JDK version.

Sample

Sample Element	Details
Unit of analysis	One Java project/build attempt under a specific JDK version.
Source/version	Sulír et al. dataset, Zenodo Version 1.1: https://zenodo.org/records/15555218, and authors' GitHub repository: https://github.com/sulir/jdk-study, commit 16a4b3d227edece0450ce1a854de3d11b640c73f. Accessed 5 October 2026.

Sample Element	Details
Size/range	20 Java projects tested across JDK 11, 17, 21 and 23.
Selection rule	The project brief requires the first 20 eligible projects in stable order (case ID, then file path, then chronological ID). As results.csv identifies projects using the name column, projects were sorted alphabetically by name and the first 20 were selected.

Selected Projects

No.	Project
1	0pandadev/nextron
2	1135/equationexploit
3	123yyh123/xiaofanshu
4	15knots/cmake4eclipse
5	1991wangliang/lorne_core
6	1and1/ioc-unit
7	1brc/nodejs
8	2no2name/hopperoptimizations
9	373675032/smart-medicine
10	3arthqu4ke/handlewrapper
11	3arthqu4ke/pingbypass-client
12	4drian3d/SignedVelocity
13	4pr0n/ripme
14	4ra1n/jar-analyzer-gui
15	4ra1n/super-xray
16	642933588/jiron-cloud
17	6tail/tyme4j

No.	Project
18	864381832/xjavafxtool-spring
19	918xj/spring-boot-seed
20	a24group/ssgwt-lib

Method, metric and evidence route

RQ1 — Buildability Pattern

Inputs	Procedure	Output / Decision
20 selected Java projects; JDK 11, 17, 21 and 23; project build tools.	Build each project using each specified JDK, using the appropriate build tool/configuration. Record each build as successful or failed and classify failures.	Buildability matrix and success rate for each JDK/build tool. Compare the resulting pattern with the original study.

RQ2 — JDK-Switch Repair

INPUTS	PROCEDURE	OUTPUT / DECISION
Projects from the sample that fail on JDK 23; JDK 21, 17 and 11.	Check whether each JDK 23 failure successfully builds on an earlier tested JDK, without changing the project code.	Number and proportion of JDK 23 failures successfully resolved by switching JDK. Compare with the original study's JDK-switch finding.

Expected result and decision rule

Outcome	Meaning
Expected result	Build success rates will vary across JDK 11, 17, 21 and 23, showing a broadly similar pattern to the original study.

Outcome	Meaning
REPRODUCED	The bounded sample shows the same main JDK/build-tool success pattern as the original study.
NOT REPRODUCED	A valid completed experiment shows a clearly different JDK/build-tool success pattern.
INCONCLUSIVE / BLOCKED	Insufficient valid builds or environment/data problems prevent a reliable comparison.

Outcome	Meaning
Expected result	Some projects failing on JDK 23 will successfully build using an earlier tested JDK without code changes.
REPRODUCED	At least one valid JDK 23 failure successfully builds using JDK 21, 17 or 11 without code changes.
NOT REPRODUCED	None of the valid JDK 23 failures successfully build using JDK 21, 17 or 11.
INCONCLUSIVE / BLOCKED	There are no valid JDK 23 failures to test, or execution/environment problems prevent a valid comparison.

Check	Evidence / result
Feasibility check	**PASS —** Dataset and GitHub artefact successfully accessed. JDK 21.0.12 confirmed working using java -version and javac -version
Environment setup	**PASS —** The authors' repository was successfully set up on Ubuntu using Docker and Python 3.11.14. All tests passed (37 tests and 36 subtests)